

Year 10 (Pathway 2 and 3) - Surface Area Investigation

Name: _____

Mark: _____/

Weighting: 10%

When: To be completed in class, week 9 or 10

Resources provided: Blocks, surface area nets to make into shapes and tape/ glue.

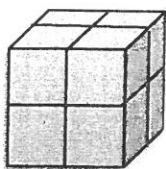
Conditions: Independent work, calculators allowed.

Total time: 1 period, extra time may be allocated at teacher discretion.

Part 1: Cube

1. Find the surface area of the 2 cm cube.

(2 marks)



$$4 \times 6 = 24 \text{ cm}^2$$

✓ ✓

2. When you join two cubes like this together the surface area changes. Find the new surface area after adding the two cubes together. Hint draw a diagram. (2 marks)

$$(24 \times 2) - 8 = 40 \text{ cm}^2$$

✓ ✓

3. Is it just the same as double the surface area of the cube? Why/ Why not?

(2 marks)

No .

Because two faces are joined / hidden. ✓

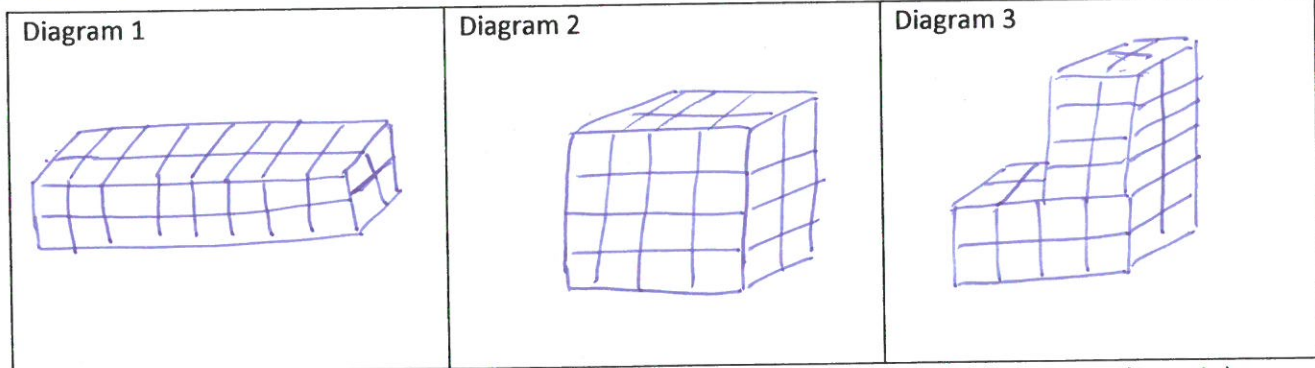
- Any reasonable explanation .

Part 2- Rectangular Prisms- Using the Rectangular Prism from Part 1.

1. Join two of the same rectangular prisms together in 3 different ways. Draw a diagram of each.

(3 marks)

* OTHERS ARE POSSIBLE .



2. Calculate the total surface area (TSA) of each of the composite shapes above. (9 marks)

TSA Calculations- Diagram 1	TSA Calculations- Diagram 2	TSA Calculations- Diagram 3
$16 \times 4 = 64$ $4 \times 2 = 8$ $64 + 8 = 72\text{cm}^2$	$8 \times 4 = 32$ $16 \times 2 = 32$ $32 + 32 = 64\text{cm}^2$	$16 \times 2 = 32$ $12 \times 1 = 12$ $8 \times 1 = 16$ $4 \times 2 = 12$ $32 + 12 + 12 + 16 = 72\text{cm}^2$

3. Are these surface areas the same? Why /Why not?

3 marks

No.

Because various faces are covered each time.

✓✓✓

Any reasonable explanation.

Part 3: Painting a ramp

1. Determine the surface area of the triangular prism. You may use the net provided to assist you.

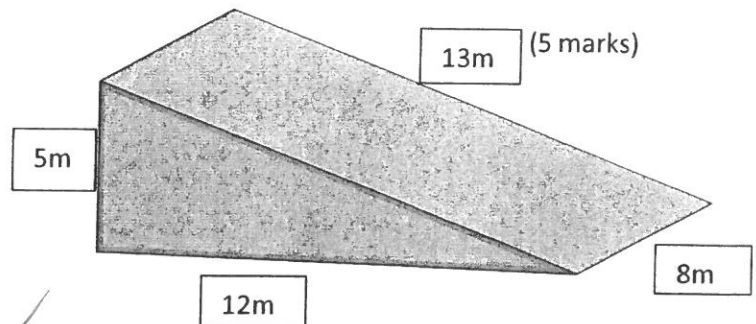
$$13 \times 8 = 104 \quad \checkmark$$

$$12 \times 5 = 60 \quad \checkmark$$

$$5 \times 8 = 40 \quad \checkmark$$

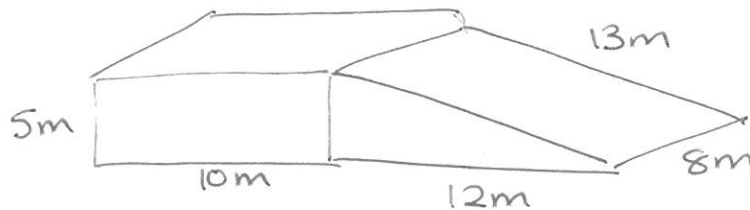
$$12 \times 8 = 96 \quad \checkmark$$

$$\text{Total SA} = 300 \text{ m}^2 \quad \checkmark$$



2. A rectangular prism is added to the end of the triangular prism the dimensions of the rectangular prism are 5m x 8m x 10m.

Draw this connected to the triangular prism. (A net is provided if required) (1 mark)



3. This combined prism is actually being used as a timber ramp that is set on the ground. The top and all of the sides need to be painted. Determine the total area to be painted so we can buy enough paint. (4 marks)

$$300 - 40 = 260$$

$$10 \times 5 \times 2 = 100$$

$$5 \times 8 = 40$$

$$10 \times 8 \times 2 = 160$$

$$\text{Total SA} = 560 \text{ m}^2$$

4. Two coats of paint are required. Each litre of paint covers 16 square metres. How many litres of paint are required? Round up to the nearest whole litre. (3 marks)

$$560 \text{ m}^2 \times 2 = 1120 \quad \checkmark$$

$$1120 \div 16 = 70 \text{ L} \quad \checkmark$$

$$\begin{aligned} 560 - (22 \times 8) &= 384 \text{ m}^2 \\ 384 \times 2 &= 768 \text{ m}^2 \\ 768 \div 16 &= 48 \text{ L} \end{aligned} \quad \checkmark$$

(Base is not painted)

